The Hillingdon Hospitals
NHS Foundation Trust

Caesarean Wound Management within Maternity

Author name and job title	Adelaide Seemann, Tissue Viability Nurse
Guideline Lead name & job title	Adelaide Seemann, Tissue Viability Nurse
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Caesarean Wound Management within Maternity

1. The Aim of this guideline is to promote a standardised approach to wound care within the Maternity department, thereby enhancing wound care practices and supporting effective wound healing. This guideline ensures that Staff at The Hillingdon Hospitals are provided with current, evidence-based knowledge regarding post-natal wounds.

2. All patients admitted to The Hillingdon Hospital are required to have a Purpose T and Body Map completed upon admission and as their condition changes. Both forms can be accessed and completed on Cerner through ADHOC and Care Facilitation.

3. Caesarean surgical incisions represent the most prevalent postnatal wound within the maternity field. Data from 2024 indicates that 1 in 4 pregnancies in the UK result in a Caesarean birth. These procedures may be planned (elective) or conducted as emergencies due to complications arising during pregnancy or labour. Regardless of the circumstances, healthcare professionals are obligated to provide women with consistent, evidence-based guidance regarding caesarean wound healing, the risk of surgical site infections (SSI) and preventive measures that can be taken to mitigate these risks.

3.1. **Primary and Secondary Intention Closure Techniques**

Primary intention is a complete closure technique that allows for more effective wound management. It results in less tissue loss, reduces the risk of infection, and minimizes the likelihood of wound dehiscence. The NICE Guidelines (2019) recommend the use of sutures over staples for skin closure following a caesarean delivery to reduce the risk of superficial wound dehiscence. Secondary intention refers to the healing process for wounds that cannot be fully closed surgically due to significant tissue loss or the inability to securely approximate the wound edges. This method of closure is associated with a prolonged healing duration (Wollheim & Meeker, 2023). To facilitate wound healing, negative pressure therapy is commonly utilised; however, this can lead to an extended length of hospital stay for patients and may have a notable impact on both their psychological and physical well-being.

3.2. **Contributing factors to wound Healing**

Contributing factors may delay wound healing and should be highlighted as potential risks for all patients with wounds. These factors include a body mass index (BMI) greater than 35, a pendulous abdomen, malnutrition, diabetes, various diseases, smoking status, low levels of albumin or haemoglobin, and a history of wound healing concerns/ issues.

3.3. **Wound Dehiscence**

Dehiscence refers to a partial or complete separation of wound edges that were previously approximated, resulting from inadequate wound healing. The depth of a dehiscence may range from superficial skin level to a deeper involvement that may affect subcutaneous tissue. In rare cases, it may also involve underlying muscle tissue. Patients identified above in Contributing factors to wound healing are at higher risk of a wound dehiscence. It is important to note that a wound dehiscence does not automatically indicate the presence of a wound infection.

4. The current NICE Recommendation (2024) advises that standard postoperative surgical dressings be removed 24 hours after a caesarean section. This practice facilitates a

comprehensive wound evaluation and promotes regular assessments of the wound. If no issues with the wound are identified, no additional dressings are necessary. Patients are encouraged to maintain the cleanliness and dryness of the area, while avoiding the use of soaps. If Single Use Negative Pressure Wound Therapy (PICO7) has been utilised, dressings should be monitored closely. However, it is recommended PICO dressings should remain in place for a full duration of 7 days unless there are specific concerns that indicate early removal.

4.1. **Removal of non-absorbable Sutures/ Staples**
NICE recommends the use of absorbable subcuticular sutures for Caesarean closure rather than staples (2019) however, due to certain risk factors patients may occasionally require closure with non-absorbable sutures or staples. Non-absorbable sutures commonly “Prolene” should be removed on day 5. Staples are to be removed on day 10. In certain cases, particularly for higher-risk patients or where there are concerns, the removal of clips may be delayed until day 14. There is no evidence to suggest removing alternate staples results in better patient outcomes, therefore if any wound dehiscence concerns arise all staples can remain in place for the maximum of 14 days.

5. Accurate **documentation** is essential in validating a wound assessment. Documentation plays a crucial role in ensuring continuity of care and providing evidence regarding the healing status of a wound. It is recommended that wound and dressing assessments be documented at every shift to facilitate ongoing review and management.

6. **Dressing Changes** are not routinely required, Usually the post operative dressing is removed after 24 hours and not reapplied. For complex non-healing sterile wounds all dressing changes must be complete using an Aseptic Non-Touch Technique, this helps to reduce the risk of infection/ isolation of infection.

6.1. Wound Cleansing should occur at each dressing change, for non-infected wounds Sodium Chloride is recommended however for infected wounds a Prontosan soak may be required.
6.2. Dressing changes are typically not required on a routine basis. Generally, the postoperative dressing can be removed after 24 hours without the need for reapplication. For complex non-healing sterile wounds, it is essential to complete all dressing changes using an Aseptic Non-Touch Technique, as this practice helps minimise the risk of infection and isolate any potential sources of infection.
6.3. Considerations for dressing selection; Ensure dressing selection ensures an adequate moisture balance. Utilise absorbent dressings to manage excess exudate. Please confirm any relevant allergies with the patient prior to applying dressings.

7. In cases where a **wound infection is suspected**, it is important to obtain swabs and submit them to the laboratory. Please note that once a swab is collected and sent for analysis, it may take up to a week for bacterial cultures to grow and the results to be released. During this time, please refrain from re-swabbing the wound. Patients who are clinically stable and asymptomatic do not require admission while awaiting swab results.

7.1. Obtaining a wound swab: Where possible specimens should be collected before starting any antimicrobial therapy.

1.Wounds should be cleaned with Sodium Chloride before obtaining a swab

2.Swab the deepest area of the wound by moving the swab across the wound surface, at the same time as rotating. Care should be taken to ensure the that the swab only encounters the wound surface, avoid touching the swab on surrounding skin.

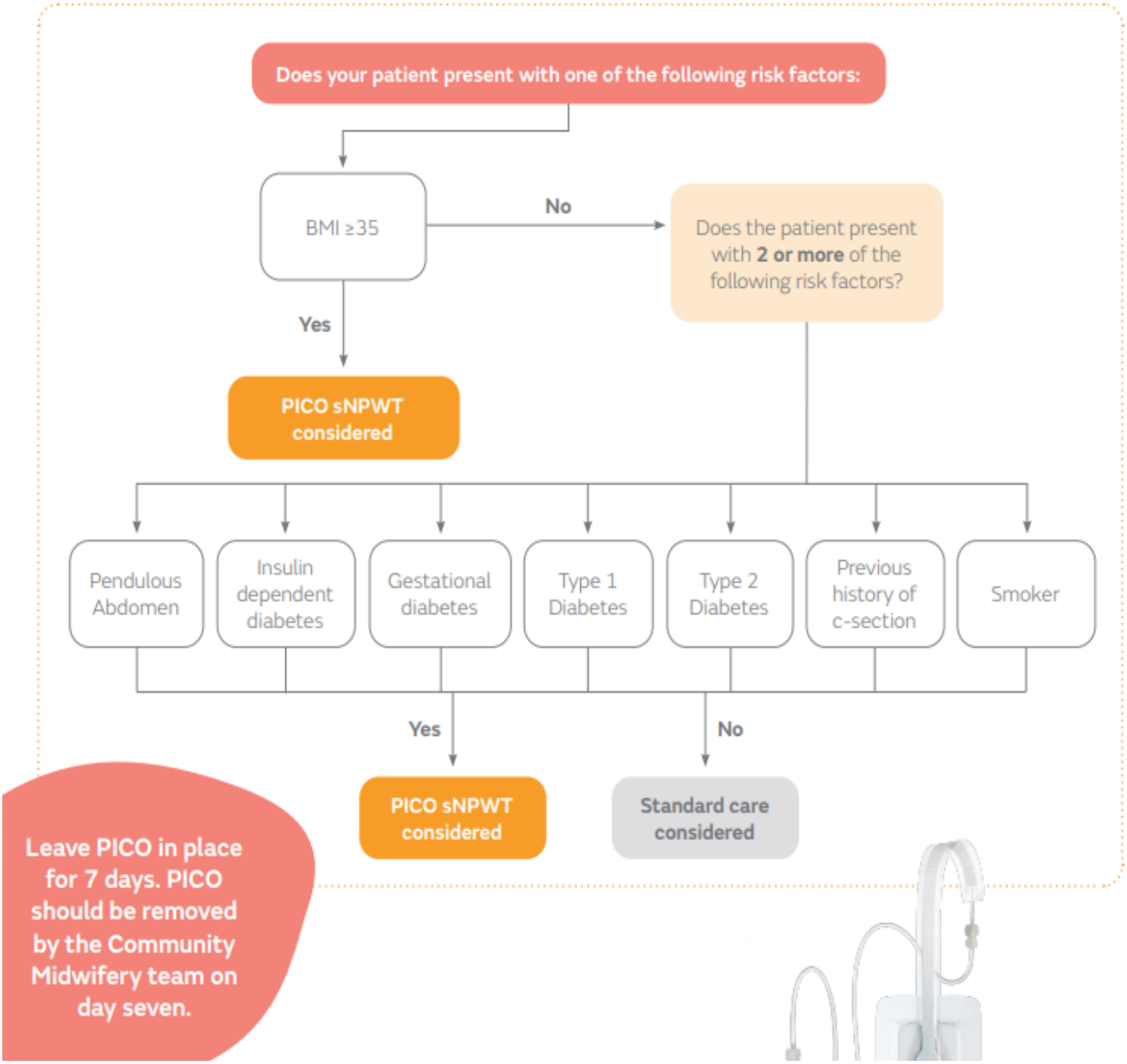
3.Immediately after collection the swab should be returned to its plastic sheath (placed into the transport medium.)

4.Transport to the laboratory as soon as possible.

If transport is delayed, refrigeration is preferable to storage at ambient temperature
When an infection is confirmed please do not re-swab wound for a minimum of 2 weeks after completion of antibiotics and only resend swabs if concerns of infection persist.

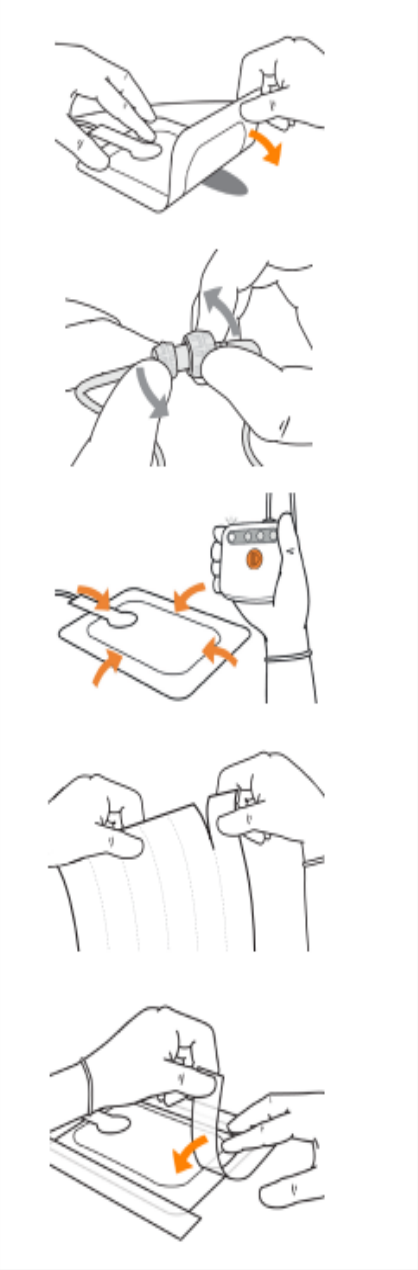
8. Single Use Negative Pressure Wound Therapy System (PICO) NICE Guidance 2019 Gold Standards.

8.1. Caesarean Section PICO Pathway



8.2. **PICO Application**

Application



Dress

1. Clean and prepare wound according to local protocol. Refer to PICO 7 Instructions for Use for details on application.
2. Peel off the central release handle and place the dressing centrally over the wound. The soft port should be uppermost from the wound (depending on the patient's primary position).
3. Remove the other two handles and smooth the dressing around the wound to prevent creasing.

Press

4. Insert the batteries into the pump. Following this all four indicators should illuminate for 3 seconds.
5. Join the pump and dressing tubes by twisting together the connectors.
6. Press the orange button to start the application of negative pressure. The green OK indicator and orange air leak indicator will start to flash together.

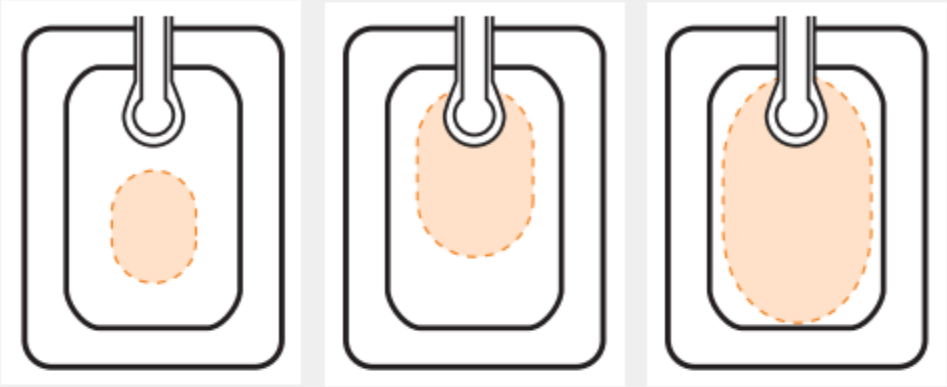
   Depending on the size of the wound, the pump should take up to 65 seconds to establish NPWT. If after 65 seconds the system has not established NPWT, just the orange air leak indicator will flash

Go

7. Apply the fixation strips all the way around the dressing border.
8. The pump has a seven-day life and the dressing may be left in place for up to seven days depending on the level of exudate.[1-3]
9. When a filler is used, the filler and the PICO dressing should be changed two to three times a week.

8.3. **PICO Dressing Change**

Dressing change



A
Dressing properly positioned and is acceptable to be left in place

B
Dressing requires change

C
Dressing requires change

If the dressing appears ready for changing (see diagrams A – C) within the seven-day life of the pump, press the orange button and disconnect the dressing from the pump.

The dressing should be disposed of as clinical waste. The batteries should be removed from the pump; and both batteries and pump disposed of according to local regulations.

Note: The soft port should be uppermost from the wound (depending on the patient's primary position).

9. Upon ***discharge from hospital*** care, all patients who have undergone a caesarean section must receive a follow up wound assessment. Patients being discharged with PICO must be provided the single spare dressing to enable a dressing change upon discharge IF required.

9.1. Provided there are no concerns regarding the wound, the patient can be followed up by the ***community midwives*** according to the relevant care pathway. Please ensure that the necessary supporting documentation is submitted to the ***Community Midwives Team*** upon discharge.

9.2. ***Mobile patients*** with complex wounds care needs or those who require care beyond the initial 28 days, please direct them to the ambulant wound clinic. Referral form is located on the Trust Intranet

9.3. ***Immobile/ Housebound*** patients with complex wound care needs; please ensure that a referral is made to the District Nurse. Referral form is located on the Trust Intranet

10. The upload of medical photographs is facilitated through Mediviewer using a trusted camera-enabled device. Medical photography plays a vital role in monitoring the progression or deterioration of wounds. Additionally, obtaining medical photographs enables remote healthcare professionals to review and provide support as needed.

10.1 Consent forms for medical photography must be completed prior to uploading any images. These forms can be accessed via the intranet.

11. Associated Guidelines, Policies and Documents

- Standard Infection Prevention and Control Precautions
- Risk Associated with Infection Prevention and Control Policy
- Prevention and Management of Pressure Ulcers in Maternity
- Perineal Trauma and Episiotomy Repair

12. References

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